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/**
 * Main class representing the ATM machine
 * Handles user interaction and coordinates system operations
 */
public class ATM {

    // Attributes
    private Bank bank;
    private Card currentCard;
    private Account currentAccount;
    private boolean isAuthenticated;

    /**
     * Starts the ATM machine
     */
    public void start() {
        // TODO: Initialize ATM interface
    }

    /**
     * Authenticates user using card and PIN
     * @param card Card inserted by user
     * @param pin Entered PIN
     * @return boolean indicating success/failure
     */
    public boolean authenticateUser(Card card, int pin) {
        return false;
    }

    /**
     * Displays available options to user
     */
    public void showMenu() {
        // TODO: Display options (withdraw, deposit, balance, etc.)
    }

    /**
     * Logs out the current user
     */
    public void logout() {
        // TODO: Reset session
    }
}

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/**
 * Represents a bank system
 * Manages accounts and validates users
 */
class Bank {

    // Attributes
    private String bankName;

    /**
     * Validates card and PIN
     * @param card Card object
     * @param pin PIN number
     * @return Account if valid, otherwise null
     */
    public Account authenticate(Card card, int pin) {
        return null;
    }

    /**
     * Retrieves account by account number
     * @param accountNumber Account number
     * @return Account object
     */
    public Account getAccount(int accountNumber) {
        return null;
    }
}

/**
 * Represents a user's bank account
 */
class Account {

    // Attributes
    private int accountNumber;
    private double balance;
    private String accountHolderName;

    /**
     * Deposits money into account
     * @param amount Amount to deposit
     * @return updated balance
     */

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public double deposit(double amount) {
return 0.0;
}

/**
 * Withdraws money from account
 * @param amount Amount to withdraw
 * @return boolean indicating success/failure
 */
public boolean withdraw(double amount) {
return false;
}

/**
 * Gets current account balance
 * @return balance
 */
public double getBalance() {
return balance;
}
}

/**
 * Represents a debit/ATM card
 */
class Card {

// Attributes
private String cardNumber;
private int pin;
private int accountNumber;

/**
 * Validates entered PIN
 * @param enteredPin PIN entered by user
 * @return boolean indicating match
 */
public boolean validatePin(int enteredPin) {
return false;
}

/**
 * Gets linked account number
 * @return account number

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*/
public int getAccountNumber() {
return accountNumber;
}
}

/**
 * Abstract class representing a transaction
 */
abstract class Transaction {

// Attributes
protected Account account;
protected double amount;

/**
 * Constructor
 */
public Transaction(Account account, double amount) {
this.account = account;
this.amount = amount;
}

/**
 * Executes the transaction
 * @return boolean indicating success/failure
 */
public abstract boolean execute();
}

/**
 * Handles withdrawal transactions
 */
class WithdrawTransaction extends Transaction {

/**
 * Constructor
 */
public WithdrawTransaction(Account account, double amount) {
super(account, amount);
}

/**
 * Executes withdrawal

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*/
@Override
public boolean execute() {
return false;
}
}

/**
 * Handles deposit transactions
 */
class DepositTransaction extends Transaction {

/**
 * Constructor
 */
public DepositTransaction(Account account, double amount) {
super(account, amount);
}

/**
 * Executes deposit
 */
@Override
public boolean execute() {
return false;
}
}

/**
 * Handles balance inquiry transactions
 */
class BalanceInquiryTransaction extends Transaction {

/**
 * Constructor
 */
public BalanceInquiryTransaction(Account account) {
super(account, 0);
}

/**
 * Executes balance check
 */
@Override

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public boolean execute() {
    return false;
}

/**
 * Returns account balance
 * @return balance
 */
public double getBalance() {
    return 0.0;
}
}

/**
 * Represents the ATM's cash dispenser
 */
class CashDispenser {

    // Attributes
    private int cashAvailable;

    /**
     * Dispenses cash
     * @param amount Amount to dispense
     * @return boolean indicating success/failure
     */
    public boolean dispenseCash(double amount) {
        return false;
    }

    /**
     * Checks if enough cash is available
     * @param amount requested amount
     * @return boolean
     */
    public boolean hasSufficientCash(double amount) {
        return false;
    }
}

/**
 * Represents the ATM's deposit slot
 */
class DepositSlot {

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/**  
 * Accepts deposit  
 * @return boolean indicating if deposit was successful  
 */  
public boolean acceptDeposit() {  
    return false;  
}  
}
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